

and bottom-up information; the constraints are applied during training to ensure structural consistency; and the enhanced loss function that governs its optimization, balancing accuracy across all hierarchical levels. These advancements make BiHRNN a more robust and versatile tool for forecasting in complex hierarchical datasets, particularly in domains like inflation modeling where cross-level interactions are critical.

3.2.1 Bidirectional Information Flow

BiHRNN is formulated as a risk minimization optimization problem. Instead of HRNN's informative prior, BiHRNN introduces two constraints on the models parameters. One ties the parameters of each time series to its parent's (similar to HRNN's prior) and the second ties the parameters of each of its' child series, with an appropriate weight:

- **Parent-Node Constraint:** This constraint governs the relationship between the parameters of a node n and its parent π_n . By aligning the parameters of node n with those of its parent, this constraint ensures hierarchical consistency and allows top-down information to flow through the network.
- **Child-Node Constraint:** This constraint governs the relationship between node n and its children η_{i_n} , where η_{i_n} represents the i -th child of node n . This enables the node to aggregate information from its children, allowing bottom-up feedback to influence higher-level nodes.

By incorporating these two constraints, BiHRNN enables information to flow in both directions—*downward* from parent to child and *upward* from child to parent. This bidirectional approach significantly enhances the model's ability to capture complex dependencies and interactions across hierarchical levels, leading to improved predictive performance.

3.2.2 Customized Loss Function

The information flow in BiHRNN is governed by a *customized loss function* that balances prediction accuracy with hierarchical consistency. The loss function comprises three key components:

1. **Mean Squared Error (MSE):** The primary objective of BiHRNN is to minimize prediction errors. This is achieved using the mean squared error:

$$\text{MSE} = \frac{1}{N} \sum (y - \hat{y})^2 \quad (3.4)$$

where y represents the observed values, $\hat{y}$ denotes the predicted values, and N is the number of observations.

2. **Parent Regularization (l_{parent}):** To ensure hierarchical coherence, the model penalizes the squared Euclidean distance between the parameters of a node n and its parent π_n :

$$l_{\text{parent}} = (\theta_{\text{parent}} - \theta)^2 \quad (3.5)$$

This term enforces consistency between a node and its parent, ensuring that higher-level nodes influence lower-level nodes appropriately.

3. **Child Regularization (l_{child}):** Similarly, the model incorporates the influence of child nodes through a weighted penalty:

$$l_{\text{child}} = \sum_{i \in \text{children}} w_i (\theta - \theta_{\text{child}})^2 \quad (3.6)$$

where w_i is a weight controlling the contribution of each child. This term allows the parameters of node n to aggregate information from its children, enabling bottom-up information flow.

The final loss function combines these components, weighted by hyperparameters λ_1 and λ_2 to control the relative importance of parent and child regularization:

$$\text{Loss}_{\text{BiHRNN}} = \frac{1}{N} \sum (y - \hat{y})^2 + \lambda_1 \cdot l_{\text{parent}} + \lambda_2 \cdot l_{\text{child}} \quad (3.7)$$

3.2.3 Hyperparameter Tuning

The hyperparameters λ_1 and λ_2 play a critical role in balancing the bidirectional information flow:

- A higher λ_1 emphasizes top-down influence by prioritizing alignment with parent nodes.
- A higher λ_2 strengthens bottom-up feedback from child nodes.

Proper tuning of these hyperparameters is essential for optimizing the model's performance while maintaining hierarchical consistency. To achieve this, Optuna¹ was employed for hyperparameter tuning, utilizing its Tree-structured Parzen Estimator (TPE), a Bayesian optimization approach, to efficiently explore the hyperparameter space and ensure robust performance.

3.2.4 Fixed Constraints During Training

A key feature of the BiHRNN model is its use of *fixed constraints* for the parent and child relationships throughout the training process. The procedure involves the following steps:

¹ <https://optuna.org/>

1. **Pretraining the Base Model:** The HRNN (or a similar baseline model) is first trained independently for each category, and the learned weights are saved. These weights serve as the initial representations of the parent and child nodes.
2. **Freezing the Weights:** Once the HRNN weights are trained, they are frozen and used as fixed constraints for the bidirectional model. This means that the weights representing parent and child relationships remain constant during the BiHRNN training process.
3. **Stabilization and Regularization:** By leveraging frozen weights, the BiHRNN anchors predictions, ensuring that hierarchical relationships are preserved and reducing the risk of overfitting. This approach is particularly beneficial for datasets with limited samples or high variability, as it provides a stable foundation for the bidirectional model.

3.2.5 *Summary*

The BiHRNN introduces a bidirectional approach to hierarchical modeling, leveraging dual constraints and a customized loss function to enable efficient information flow between nodes. By using fixed constraints and balancing top-down and bottom-up interactions, the model achieves superior forecasting accuracy and hierarchical coherence, establishing itself as a robust framework for hierarchical time series forecasting.

Figure 3.3 below depicts the BiHRNN architecture.